Research Proposal Title

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13th October 2026



A proposal submitted to the Faculty of Science, University of the Witwatersrand, Johannesburg, in partial fulfilment of the requirements for the degree of Bachelor of Science with Honours in Computer Science.

Abstract

[Summarise the problem, motivation, working research question, proposed approach and intended contribution. Describe expected outcomes as hypotheses rather than completed results. Write this after drafting the body.]

Declaration

[Insert the declaration required by your school for an honours research proposal. Confirm the wording with your supervisor. This template does not reproduce the reference author's doctoral-thesis declaration or signature.]

Signature

Date

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Chapter 1

Introduction

[Introduce the research problem and explain why it matters. Move from the wider context to the specific interaction you intend to investigate.]

[Describe the motivation and evidence for the problem. Distinguish reported literature findings, illustrative examples and observations you have personally reproduced.]

[State the working research question, the possible contribution and the boundaries of the project. Explain what remains unresolved.]

The remainder of this proposal is organised as follows. Chapter 2 reviews the foundations and related work. Chapter 3 states the research objectives and hypotheses. Chapter 4 describes the proposed methodology. Chapter 5 presents the research plan, success criteria and possible challenges. Chapter 6 concludes the proposal.

Chapter 2

Background and Related Work

[Establish the concepts needed to understand the research question. Compare the closest existing methods and identify the unresolved question. Rename or remove the suggested sections as the literature review develops.]

2.1 Reinforcement Learning

[Define the learning problem, policies, returns and relevant assumptions. Introduce notation consistently.]

[Citation examples: Sutton and Barto (2018); [Seo et al. 2025]. Replace this line with your literature discussion.]

2.2 Safe Reinforcement Learning and Safety Filtering

[Define safety, constraints, intervention and backup behaviour. Explain the distinction between measured safety and guarantees under assumptions.]

2.3 World Models and Learned Safety Assessment

[Review representations, predictive dynamics, safety values and uncertainty. Explain how model error affects safety assessment.]

2.4 Evolutionary Algorithms and Policy Adaptation

[Review the relevant search methods, candidate representations, variation and selection. Explain evaluation and interaction costs.]

2.5 Learning in the Presence of Safety Intervention

[Review policy/filter interaction and existing adaptation remedies. Explain the information and action-history semantics used by each method.]

2.6 Research Gap

[Synthesize the closest work into a specific remaining question. Support novelty claims with primary literature and state the limits of the search.]

Chapter 3

Research Objectives

3.1 Research Goals

[State the main research question and list concrete objectives that would answer it. Distinguish scientific objectives from implementation tasks.]

1. *[First research objective.]*
2. *[Second research objective.]*
3. *[Third research objective, if needed.]*

3.2 Research Hypotheses

[For each hypothesis, specify the expected relationship, the conditions under which it is claimed and the observation that would challenge it. Do not assume that evolutionary adaptation will outperform the alternatives.]

- *[First testable hypothesis.]*
- *[Second testable hypothesis, if needed.]*

Chapter 4

Research Methodology

[Explain how the proposed work answers the research question. The sections below are editable prompts, not decisions about the environment, world model, search space or evolutionary algorithm.]

4.1 Problem Formulation

[Define the task, observations, action interfaces, safety outcomes and adaptation objective. State assumptions and the scope of any guarantees.]

4.2 Environment and Existing Assets

[Justify the selected testbed after the literature and feasibility review. Record the policy, model and filter assets, versions and verified runtime.]

4.3 Proposed Adaptation Approach

[Describe what changes, what remains fixed, the candidate representation and the evaluation process. Justify the design choices.]

4.4 Baselines and Ablations

[Specify comparisons that isolate the contribution. Match information, behavioural representations and interaction budgets where applicable.]

4.5 Evaluation and Analysis

[Define completion, physical violations and unfinished episodes. Specify splits, repetitions, uncertainty estimates and tests of the hypotheses. Account for simulator interactions, model queries and elapsed time separately.]

4.6 Reproducibility

[Describe seeds, configurations, checkpoints, logging, action semantics and procedures for reproducing the comparisons.]

Chapter 5

Research Plan

5.1 Time Plan

[Replace the planning skeleton with agreed dates and realistic dependencies. The rows are editable milestones, not a committed schedule.]

Milestone	Planned period	Deliverable
Foundations and literature	To be agreed	Focused literature review
Research formulation	To be agreed	Question and justified proposal
Feasibility investigation	To be agreed	Reproduced baseline and diagnosis
Method investigation	To be agreed	Implemented candidate approach
Evaluation and analysis	To be agreed	Comparisons and interpreted results
Writing and revision	To be agreed	Honours research submission

Table 5.1: Provisional research milestones.

5.2 Success Criteria

[Define what evidence would answer the research question. Separate successful implementation, scientific findings and publication ambitions. A negative finding may still answer a well-formulated question.]

5.3 Possible Challenges

[Identify the material risks, including unavailable assets, insufficient task headroom, model error, evaluation cost and uncertain safety estimates. Describe feasible responses and limitations.]

Chapter 6

Conclusion

[Restate the problem, working research question and intended contribution. Summarise why the proposed investigation is justified and feasible.]

[State the scope, remaining uncertainties and expected value of the findings. Avoid presenting proposed methods or anticipated outcomes as completed results.]

References

Seo, J., K. Nakamura and A. Bajcsy (2025). *Uncertainty-aware Latent Safety Filters for Avoiding Out-of-Distribution Failures*. arXiv: 2505.00779 [cs.RO]. URL: https://arxiv.org/abs/2505.00779.

Sutton, R. S. and A. G. Barto (2018). *Reinforcement Learning: An Introduction*. 2nd ed. MIT Press.